Category:

Crypto

Name:

C3: unf0given

Message:

Description:

The ransomware operator is breaching an important file of a company. It is being kept secret using a password which can be cracked using the cryptographic algorithm that is in a text file.

Details:

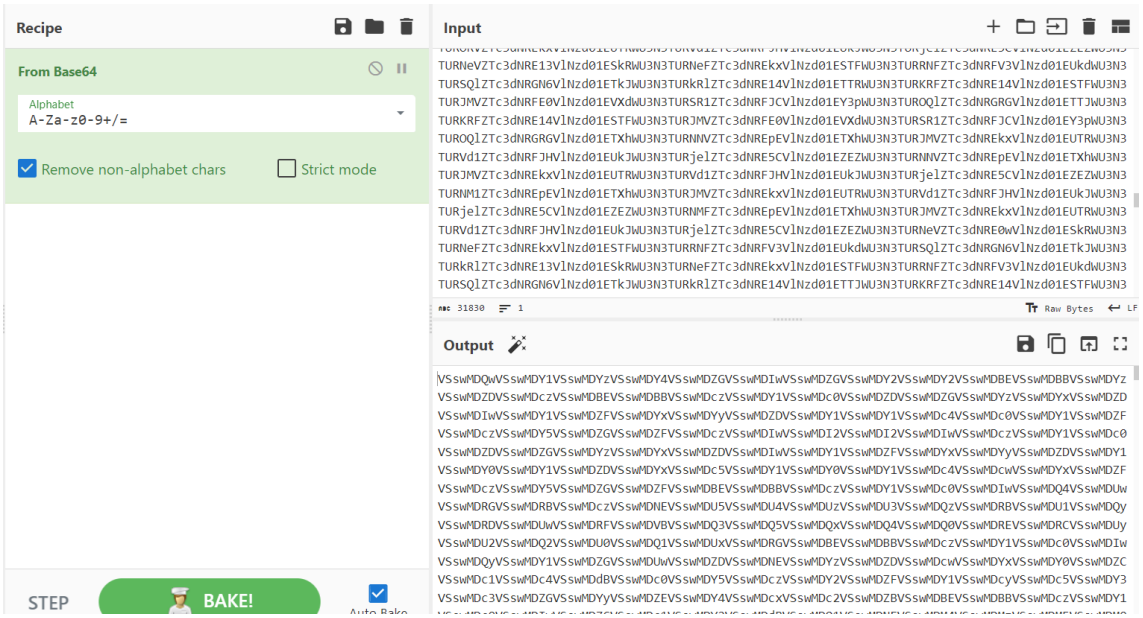
Below is a ZIP file “DoYouThinkImFragile.zip” that contains a text file and a password-protected zip file. Inside the zip file is a flag. However, the challenge is to determine the password for the zip file, which is hidden within the contents of the text file. To obtain the password for the zip file, you must successfully un-obfuscate the code or information embedded within the text file.

Instructions:

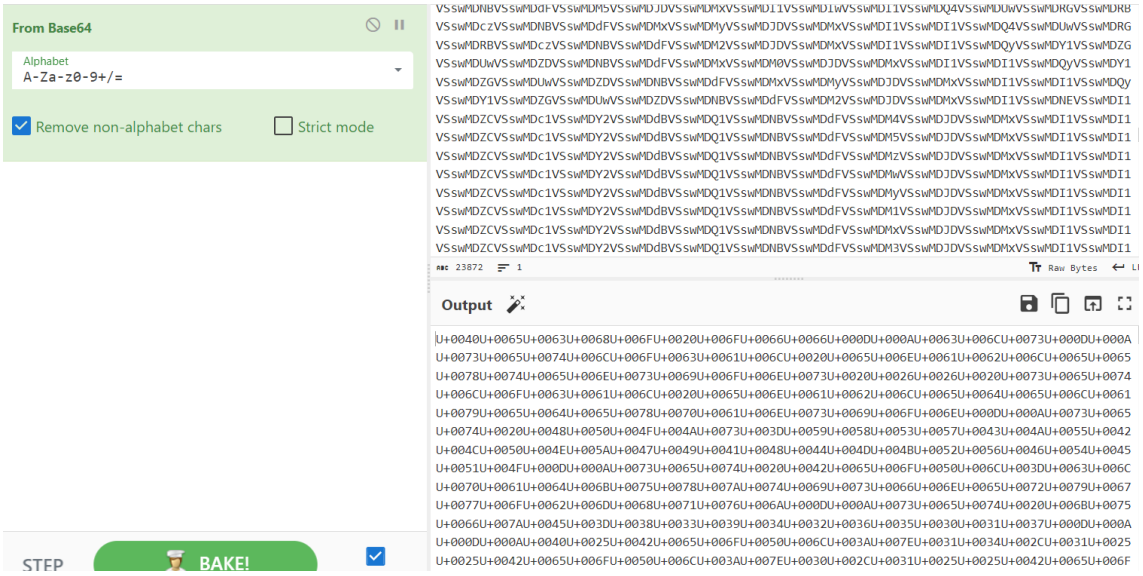
Given the text file:


```
VlNzd01EUXdWU3N3TURZMVZTc3dNRFl6VlNzd01EWTRWU3N3TURaR1ZTc3dNREl3VlNzd01EWkdWU3N3TURZMlZTc3dNRFkyVlNzd01EQkVWU3N3TUR
CQlZTc3dNRFl6VlNzd01EWkRWU3N3TURjelZTc3dNREJFVlNzd01EQkJWU3N3TURjelZTc3dNRFkxVlNzd01EYzBWU3N3TURaRFZTc3dNRFpHVlNzd0
1EWXpWU3N3TURZeFZTc3dNRFpEVlNzd01ESXdWU3N3TURZMVZTc3dNRFpGVlNzd01EWXhWU3N3TURZeVZTc3dNRFpEVlNzd01EWTFWU3N3TURZMVZTc
3dNRGM0VlNzd01EYzBWU3N3TURZMVZTc3dNRFpGVlNzd01EY3pWU3N3TURZNVZTc3dNRFpHVlNzd01EWkZWU3N3TURjelZTc3dNREl3VlNzd01ESTJW
U3N3TURJMlZTc3dNREl3VlNzd01EY3pWU3N3TURZMVZTc3dNRGMwVlNzd01EWkRWU3N3TURaR1ZTc3dNRFl6VlNzd01EWXhWU3N3TURaRFZTc3dNREl
3VlNzd01EWTFWU3N3TURaRlZTc3dNRFl4VlNzd01EWXlWU3N3TURaRFZTc3dNRFkxVlNzd01EWTBWU3N3TURZMVZTc3dNRFpEVlNzd01EWXhWU3N3TU
RjNVZTc3dNRFkxVlNzd01EWTBWU3N3TURZMVZTc3dNRGM0VlNzd01EY3dWU3N3TURZeFZTc3dNRFpGVlNzd01EY3pWU3N3TURZNVZTc3dNRFpHVlNzd
01EWkZWU3N3TURCRVZTc3dNREJCVlNzd01EY3pWU3N3TURZMVZTc3dNRGMwVlNzd01ESXdWU3N3TURRNFZTc3dNRFV3VlNzd01EUkdWU3N3TURSQlZT
c3dNRGN6VlNzd01ETkVWU3N3TURVNVZTc3dNRFU0VlNzd01EVXpWU3N3TURVM1ZTc3dNRFF6VlNzd01EUkJWU3N3TURVMVZTc3dNRFF5VlNzd01EUkR
WU3N3TURVd1ZTc3dNRFJGVlNzd01EVkJWU3N3TURRM1ZTc3dNRFE1VlNzd01EUXhWU3N3TURRNFZTc3dNRFEwVlNzd01EUkVWU3N3TURSQ1ZTc3dNRF
V5VlNzd01EVTJWU3N3TURRMlZTc3dNRFUwVlNzd01EUTFWU3N3TURVeFZTc3dNRFJHVlNzd01EQkVWU3N3TURCQlZTc3dNRGN6VlNzd01EWTFWU3N3T
URjMFZTc3dNREl3VlNzd01EUXlWU3N3TURZMVZTc3dNRFpHVlNzd01EVXdWU3N3TURaRFZTc3dNRE5FVlNzd01EWXpWU3N3TURaRFZTc3dNRGN3VlNz
d01EWXhWU3N3TURZMFZTc3dNRFpDVlNzd01EYzFWU3N3TURjNFZTc3dNRGRCVlNzd01EYzBWU3N3TURZNVZTc3dNRGN6VlNzd01EWTJWU3N3TURaRlZ
Tc3dNRFkxVlNzd01EY3lWU3N3TURjNVZTc3dNRFkzVlNzd01EYzNWU3N3TURaR1ZTc3dNRFl5VlNzd01EWkVWU3N3TURZNFZTc3dNRGN4VlNzd01EYz
JWU3N3TURaQlZTc3dNREJFVlNzd01EQkJWU3N3TURjelZTc3dNRFkxVlNzd01EYzBWU3N3TURJd1ZTc3dNRFpDVlNzd01EYzFWU3N3TURZMlZTc3dNR
GRCVlNzd01EUTFWU3N3TURORVZTc3dNRE00VlNzd01ETXpWU3N3TURNNVZTc3dNRE0wVlNzd01ETXlWU3N3TURNMlZTc3dNRE0xVlNzd01ETXdWU3N3
TURNeFZTc3dNRE0zVlNzd01EQkVWU3N3TURCQlZTc3dNREJFVlNzd01EQkJWU3N3TURRd1ZTc3dNRFkxVlNzd01EUXlWU3N3TURZMVZTc3dNRFpHVlN
zd01EVXdWU3N3TURaRFZTc3dNRE5CVlNzd01EZEZWU3N3TURNeFZTc3dNRE0wVlNzd01ESkRWU3N3TURNeFZTc3dNREkxVlNzd01ESTFWU3N3TURReV
ZTc3dNRFkxVlNzd01EWkdWU3N3TURVd1ZTc3dNRFpEVlNzd01ETkJWU3N3TURkRlZTc3dNRE15VlNzd01ETXlWU3N3TURKRFZTc3dN
RE14VlNzd01ESTFWU3N3TURJMVZTc3dNRFF5VlNzd01EWTFWU3N3TURaR1ZTc3dNRFV3VlNzd01EWkRWU3N3TUROQlZTc3dNRGRGVlNzd01ETXhWU3N
3TURNNVZTc3dNREpEVlNzd01ETXhWU3N3TURJMVZTc3dNREl3VlNzd01ESTFWU3N3TURReVZTc3dNRFkxVlNzd01EWkdWU3N3TURVd1ZTc3dNRFpEVl
Nzd01ETkJWU3N3TURkRlZTc3dNRE14VlNzd01ETTVWU3N3TURKRFZTc3dNRE14VlNzd01ESTFWU3N3TURJMVZTc3dNRFF5VlNzd01EWTFWU3N3TURaR
1ZTc3dNRFV3VlNzd01EWkRWU3N3TUROQlZTc3dNRGRGVlNzd01ETXhWU3N3TURNeVZTc3dNREpEVlNzd01ETXhWU3N3TURJMVZTc3dNREkxVlNzd01E
UXlWU3N3TURZMVZTc3dNRFpHVlNzd01EVXdWU3N3TURaRFZTc3dNRE5CVlNzd01EZEZWU3N3TURNeFZTc3dNRE15VlNzd01ESkRWU3N3TURNeFZTc3d
```

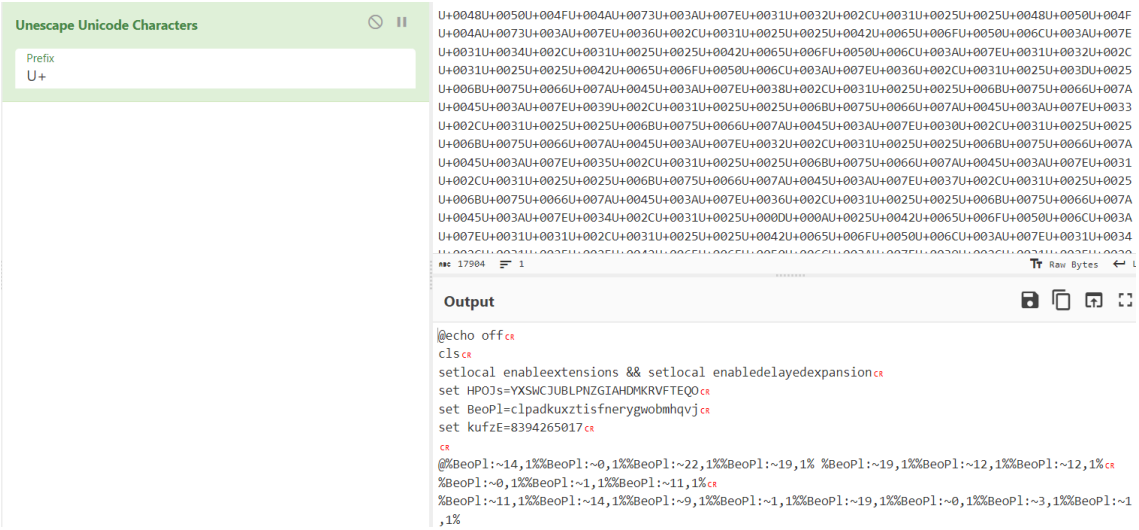
Step 1: **De-obfuscation Using base64**



Step 2: **De-obfuscation Using base64**



Step 3: **Unescape Unicode Characters (prefix: U+)**



Step 4: **String Substitution**

Each "**%[variable]:~x,y%**" represents a character from the given list below where *x* is the nth character from its variable and *y* is the number of characters to be taken from the variable. In order for the teams to get the password of the ZIP file, they need to substitute each variables into its corresponding character

1st Layer:

HPOJ		BEOPL		KUFZE	
0	Y	0	c	0	8
1	X	1	l	1	3
2	S	2	p	2	9
3	W	3	a	3	4
4	C	4	d	4	2
5	J	5	k	5	6
6	U	6	u	6	5
7	B	7	x	7	0
8	L	8	z	8	1
9	P	9	t	9	7
10	N	10	i		
11	Z	11	s		
12	G	12	f		
13	I	13	n		
14	A	14	e		
15	H	15	r		
16	D	16	y		
17	M	17	g		
18	K	18	w		
19	R	19	o		
20	V	20	b		
21	F	21	m		
22	T	22	h		
23	E	23	q		
24	Q	24	v		
25	O	25	j		

2nd Layer:

MVMWC		HOKCW		GUEFU	
0	u	0	V	0	1
1	o	1	K	1	7
2	z	2	U	2	4
3	d	3	R	3	8
4	v	4	P	4	9
5	f	5	B	5	6
6	q	6	C	6	3
7	m	7	Q	7	0
8	x	8	Y	8	5
9	w	9	D	9	2
10	l	10	H		
11	j	11	M		
12	i	12	I		
13	e	13	L		
14	t	14	F		
15	h	15	O		
16	g	16	G		
17	p	17	Z		
18	n	18	X		
19	r	19	E		
20	b	20	W		
21	y	21	A		
22	a	22	T		
23	k	23	J		
24	s	24	S		
25	c	25	N		

Obfuscated script:

```
@echo off

cls

setlocal enableextensions && setlocal enabledelayedexpansion

set HPOJs=YXSWCJUBLPNZGIAHDMKRVFTEQO

set BeoPl=clpadkuxztisfnerygwobmhqvj

set kufzE=8394265017



@%BeoPl:~14,1%%BeoPl:~0,1%%BeoPl:~22,1%%BeoPl:~19,1% %BeoPl:~19,1%%BeoPl:~12,1%%BeoPl:~12,1%

%BeoPl:~0,1%%BeoPl:~1,1%%BeoPl:~11,1%

%BeoPl:~11,1%%BeoPl:~14,1%%BeoPl:~9,1%%BeoPl:~1,1%%BeoPl:~19,1%%BeoPl:~0,1%%BeoPl:~3,1%%BeoPl:~1,1% %BeoPl:~14,1%%BeoPl:~13,1%%BeoPl:~3,1%%BeoPl:~20,

1%%BeoPl:~1,1%%BeoPl:~14,1%%BeoPl:~14,1%%BeoPl:~7,1%%BeoPl:~9,1%%BeoPl:~14,1%%BeoPl:~13,1%%BeoPl:~11,1%%BeoPl:~10,1%%BeoPl:~19,1%%BeoPl:~13,1%%Beo

Pl:~11,1%

&& %BeoPl:~11,1%%BeoPl:~14,1%%BeoPl:~9,1%%BeoPl:~1,1%%BeoPl:~19,1%%BeoPl:~0,1%%BeoPl:~3,1%%BeoPl:~1,1% %BeoPl:~14,1%%BeoPl:~13,1%%BeoPl:~3,1%%BeoPl:~

20,1%%BeoPl:~1,1%%BeoPl:~14,1%%BeoPl:~4,1%%BeoPl:~14,1%%BeoPl:~1,1%%BeoPl:~3,1%%BeoPl:~16,1%%BeoPl:~14,1%%BeoPl:~4,1%%BeoPl:~14,1%%BeoPl:~7,1%%BeoP

l:~2,1%%BeoPl:~3,1%%BeoPl:~13,1%%BeoPl:~11,1%%BeoPl:~10,1%%BeoPl:~19,1%%BeoPl:~13,1%

%BeoPl:~11,1%%BeoPl:~14,1%%BeoPl:~9,1% %HPOJs:~17,1%%HPOJs:~20,1%%HPOJs:~17,1%%BeoPl:~18,1%%BeoPl:~0,1%=%BeoPl:~6,1%%BeoPl:~19,1%%BeoPl:~8,1%%Beo

Pl:~4,1%%BeoPl:~24,1%%BeoPl:~12,1%%BeoPl:~23,1%%BeoPl:~21,1%%BeoPl:~7,1%%BeoPl:~18,1%%BeoPl:~1,1%%BeoPl:~25,1%%BeoPl:~10,1%%BeoPl:~14,1%%BeoPl:~9,1%

%BeoPl:~22,1%%BeoPl:~17,1%%BeoPl:~2,1%%BeoPl:~13,1%%BeoPl:~15,1%%BeoPl:~20,1%%BeoPl:~16,1%%BeoPl:~3,1%%BeoPl:~5,1%%BeoPl:~11,1%%BeoPl:~0,1%

%BeoPl:~11,1%%BeoPl:~14,1%%BeoPl:~9,1% %HPOJs:~12,1%%HPOJs:~6,1%%BeoPl:~14,1%%BeoPl:~12,1%%BeoPl:~6,1%=%kufzE:~8,1%%kufzE:~9,1%%kufzE:~3,1%%kufzE:~0

,1%%kufzE:~2,1%%kufzE:~5,1%%kufzE:~1,1%%kufzE:~7,1%%kufzE:~6,1%%kufzE:~4,1%

%BeoPl:~11,1%%BeoPl:~14,1%%BeoPl:~9,1% %BeoPl:~22,1%%HPOJs:~25,1%%HPOJs:~18,1%%HPOJs:~4,1%%HPOJs:~3,1%=%HPOJs:~20,1%%HPOJs:~18,1%%HPOJs:~6,1%%

HPOJs:~19,1%%HPOJs:~9,1%%HPOJs:~7,1%%HPOJs:~4,1%%HPOJs:~24,1%%HPOJs:~0,1%%HPOJs:~16,1%%HPOJs:~15,1%%HPOJs:~17,1%%HPOJs:~13,1%%HPOJs:~8,1%%

HPOJs:~21,1%%HPOJs:~25,1%%HPOJs:~12,1%%HPOJs:~11,1%%HPOJs:~1,1%%HPOJs:~23,1%%HPOJs:~3,1%%HPOJs:~14,1%%HPOJs:~22,1%%HPOJs:~5,1%%HPOJs:~2,1%

%HPOJs:~10,1%

%BeoPl:~11,1%%BeoPl:~14,1%%BeoPl:~9,1% %HPOJs:~0,1%%BeoPl:~0,1%%kufzE:~0,1%=echo %%hOKCW:~16,1%%%%MVMwc:~1,1%%%%MVMwc:~1,1%%%%MVMwc:~3,1%

% %%MVMwc:~11,1%%%%MVMwc:~1,1%%%%MVMwc:~20,1%% %%MVMwc:~22,1%%%%MVMwc:~18,1%%%%MVMwc:~3,1%% %%hOKCW:~6,1%%%%MVMwc:~1,1%%%%MV

Mwc:~18,1%%%%MVMwc:~16,1%%%%MVMwc:~19,1%%%%MVMwc:~22,1%%%%MVMwc:~14,1%%%%MVMwc:~0,1%%%%MVMwc:~10,1%%%%MVMwc:~22,1%%%%MVMwc:~1

4,1%%%%MVMwc:~12,1%%%%MVMwc:~1,1%%%%MVMwc:~18,1%%%%MVMwc:~24,1%%... %%HPOJs:~22,1%%%%MVMwc:~1,1%% %%MVMwc:~1,1%%%%MVMwc:~17,1%%%

%MVMwc:~13,1%%%%MVMwc:~18,1%% %%MVMwc:~14,1%%%%MVMwc:~15,1%%%%MVMwc:~13,1%% %%hOKCW:~17,1%%%%hOKCW:~12,1%%%%hOKCW:~4,1%% %%MV

Mwc:~5,1%%%%MVMwc:~12,1%%%%MVMwc:~10,1%%%%MVMwc:~13,1%%, %%MVMwc:~15,1%%%%MVMwc:~13,1%%%%MVMwc:~19,1%%%%MVMwc:~13,1%% %%MVMwc:~

12,1%%%%MVMwc:~24,1%% %%MVMwc:~14,1%%%%MVMwc:~15,1%%%%MVMwc:~13,1%% %%MVMwc:~17,1%%%%MVMwc:~22,1%%%%MVMwc:~24,1%%%%MVMwc:~24,1%

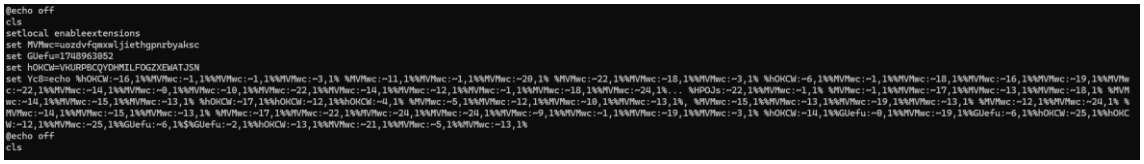
%%%MVMwc:~9,1%%%%MVMwc:~1,1%%%%MVMwc:~19,1%%%%MVMwc:~3,1%% %%hOKCW:~14,1%%%%GUefu:~0,1%%%%MVMwc:~19,1%%%%GUefu:~6,1%%%%hOKCW:

~25,1%%%%hOKCW:~12,1%%%%MVMwc:~25,1%%%%GUefu:~6,1%%$%%GUefu:~2,1%%%%hOKCW:~13,1%%%%MVMwc:~21,1%%%%MVMwc:~5,1%%%%MVMwc:~13,1%%

@%BeoPl:~14,1%%BeoPl:~0,1%%BeoPl:~22,1%%BeoPl:~19,1% %BeoPl:~19,1%%BeoPl:~12,1%%BeoPl:~12,1%

%BeoPl:~0,1%%BeoPl:~1,1%%BeoPl:~11,1%
```

You can make it readable by adding "echo " to the beginning of each line and running it as a batch file.



However, line 7 is still obfuscated. Let's remove last 2 lines and “Yc8=” in the beginning of line 7, and run it again as a batch file:
(Since the variable “HPOJs” is also used in line 7, you need to insert the definition of this variable.)

```
@echo off

cls

setlocal enableextensions

set HPOJs=YXSWCJUBLPNZGIAHDMKRVFTEQO

set MVMwc=uozdvfqmxwljiethgpnrbyaksc

set GUefu=1748963052

set hOKCW=VKURPBCQYDHMILFOGZXEWATJSN

echo %hOKCW:~16,1%%MVMwc:~1,1%%MVMwc:~1,1%%MVMwc:~3,1% %MVMwc:~11,1%%MVMwc:~1,1%%MVMwc:~20,1% %MVMwc:~22,1%%MVMwc:~18,1%%MVMwc:~3,1%

%hOKCW:~6,1%%MVMwc:~1,1%%MVMwc:~18,1%%MVMwc:~16,1%%MVMwc:~19,1%%MVMwc:~22,1%%MVMwc:~14,1%%MVMwc:~0,1%%MVMwc:~10,1%%MVMwc:~22,1%%MV

Mwc:~14,1%%MVMwc:~12,1%%MVMwc:~1,1%%MVMwc:~18,1%%MVMwc:~24,1%... %HPOJs:~22,1%%MVMwc:~1,1% %MVMwc:~1,1%%MVMwc:~17,1%%MVMwc:~13,1%%MVMw

c:~18,1% %MVMwc:~14,1%%MVMwc:~15,1%%MVMwc:~13,1% %hOKCW:~17,1%%hOKCW:~12,1%%hOKCW:~4,1% %MVMwc:~5,1%%MVMwc:~12,1%%MVMwc:~10,1%%MVMwc:

~13,1%, %MVMwc:~15,1%%MVMwc:~13,1%%MVMwc:~19,1%%MVMwc:~13,1% %MVMwc:~12,1%%MVMwc:~24,1% %MVMwc:~14,1%%MVMwc:~15,1%%MVMwc:~13,1% %MVMwc

:~17,1%%MVMwc:~22,1%%MVMwc:~24,1%%MVMwc:~24,1%%MVMwc:~9,1%%MVMwc:~1,1%%MVMwc:~19,1%%MVMwc:~3,1% %hOKCW:~14,1%%GUefu:~0,1%%MVMwc:~19,1%

%GUefu:~6,1%%hOKCW:~25,1%%hOKCW:~12,1%%MVMwc:~25,1%%GUefu:~6,1%$%GUefu:~2,1%%hOKCW:~13,1%%MVMwc:~21,1%%MVMwc:~5,1%%MVMwc:~13,1%
```

Output of the above script:

Good job and Congratulations... To open the ZIP file, here is the password

F1r3Nlc3$4Lyfe

You can extract “Congratulations!.png” file with the above password.

